NICHOLAS PREVITALI

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AI Engineering Lead | Production RAG, Multi-Agent Systems & LLM Evaluation

AI Engineering Lead with 5 years of experience architecting and deploying production-grade AI systems for Fortune Global 500 clients. Built and led a team of 5+ engineers delivering 10+ solutions across legal, e-commerce, retail, and HR domains. Hands-on expert in grounded RAG on specialized corpora (including ~100K Italian legal documents), hybrid retrieval, multi-agent orchestration (LangGraph, LangChain, MCP), LLM fine-tuning and systematic evaluation, deployed on Azure. Track record of 91% retrieval recall@10 on legal RAG and eliminating manual workflows by up to 100%.

Work Experience

S.I. 2001 SpA, Bergamo

Head of AI Business Unit | Jan 2023 – Present

Leading the AI division end-to-end: strategy, architecture, delivery, and client management across multiple concurrent enterprise projects.

- **Architected Legal RAG platform** for 5 law firms processing ~100K Italian legal documents (rulings, regulations, statutes). Cross-citation handling via hybrid retrieval combining vector store with LLM-constructed knowledge graph. 91% retrieval recall@10, 89% answer relevance evaluated via LLM-as-judge and legal expert validation. Included automated document generation pipeline.
- **Architected and currently deploying an 8-agent multi-agent platform for NTT** (Fortune Global 500) for retail planogram optimization. Each agent handles a specific domain responsibility (planogram analysis, product placement, shelf optimization, reporting). Built on RAG with a fine-tuned LLM for planogram domain comprehension, deployed on Azure. Includes full RAG evaluation pipeline.
- **Built multilingual HR AI assistant for Jakala** (2,000+ employees, scalable to 12,000) supporting IT, EN, ES, FR, DE. Handles ~7,000 daily requests across 150+ HR topics (PTO, room booking, company policies, newsletters). Hybrid search (semantic + keyword) with retroactive fine-tuning based on tester feedback. Intelligent topic routing: user queries map to pre-configured topics that navigate to correct app/web pages.
- **Designed GEO/SEO optimization platform** for beauty and fashion e-commerce: automated product descriptions, FAQs, structured data (JSON-LD), and multilingual tone-of-voice adaptation in up to 6 languages. Reduced content optimization time by ~80%. Validated AI-generated content discoverability via Google Search Console and GPT shopping domain prompt testing.
- **Built multi-tenant recommendation + RAG platform for charity sector:** non-profit organizations discover relevant projects via personalized recommendations and conversational RAG interface. Innovative two-tower-inspired architecture using vector DB instead of traditional serving infrastructure for cost optimization. Recommendations evolve dynamically based on user behavior patterns. Sub-100ms response times in production.
- **Architected MCP-based developer productivity platform** integrating Confluence, Slack, and Jira as Single Source of Truth for AI coding assistants. Custom agents with single-scope responsibilities (code review, documentation ingestion, context building) orchestrating multi-model cooperation (Claude + Gemini). Used daily by ~15 developers as the shared context layer for AI-assisted coding.
- Designed reliability layers across all LLM systems: guardrails, fallback policies, tool-call constraints, structured outputs, human-in-the-loop escalation, and tracing of latency, cost and hallucination rate (Langfuse).

Senior AI Engineer & Team Lead | Apr 2022 – Dec 2022

Managed 5+ engineers while maintaining hands-on architecture and delivery.

- **Built CNN web page classifier** for e-commerce SEO: full ML pipeline (data collection, labeling, training, validation, monitoring) achieving 98% accuracy and F1 0.95. Eliminated 100% of manual classification across thousands of web pages, enabling downstream SEO automation.
- Delivered corporate AI training to ~50 employees. Scaled team from scratch, establishing Agile/Scrum workflows and delivery standards for enterprise clients.

AI Engineer | Apr 2021 – Mar 2022

First AI hire. Built and fine-tuned CNN-based computer vision models (image classification, object detection, similarity search) using transfer learning. Integrated AI workflows into production using FastAPI microservices, Docker, and Azure AKS with load balancing, caching, and distributed inference. Built monitoring pipelines tracking latency, throughput, model accuracy, and failure modes. Translated business processes into automated ML workflows by collaborating with operations, HR, finance, and technical teams.

Technical Skills

AI & LLMs: LangChain, LangGraph, MCP, Azure OpenAI (GPT-5 / GPT-6 family), Claude, Gemini • Multi-Agent Orchestration, Guardrails, HITL • Prompt Engineering, Function Calling, Structured Outputs

RAG & Search: Hybrid Search (Dense + BM25, RRF), Reranking, Knowledge Graphs • pgvector, Supabase, Pinecone, Qdrant, Weaviate • Docling, Document Ingestion & Chunking

Evaluation: RAGAS (faithfulness, answer relevance, context precision/recall) • Recall@k, MRR, nDCG • LLM-as-Judge, Golden Sets with Domain Experts, Regression Testing

ML & CV: CNNs, Transfer Learning • LLM Fine-tuning (OpenAI models, Flan-T5) • PyTorch, TensorFlow • Recommendation Systems (Two-Tower, Collaborative Filtering)

AI-Assisted Engineering: Claude Code (daily use in delivery) • Plugin development: skills, subagents, hooks, slash commands • MCP integration

Infra: Python, FastAPI • Docker, Kubernetes (Azure AKS) • Azure Cloud • CI/CD, Git • LLMOps, Observability & Tracing (Langfuse)

Education

MSc Computer Engineering, Summa Cum Laude (110/110) | Università di Bergamo | 2019–2022
Thesis: *Extraction of Semantic Topological Maps from 2D Occupational Maps.* Co-authored with C. Celestino, B. Caglioni, D. Brugali.

BSc Computer Engineering | Università di Bergamo | 2016–2019

Research & Open Source

LLM-powered recommendation and agentic retrieval: 50+ papers studied (neural CF, retrieval-augmented recommendations).

Turbocharge (github.com/nicodiansk/turbocharge): open-source Claude Code plugin I built and published (10 skills, 6 agents, 2 hooks): brainstorm → plan → build → review pipeline for test-driven AI-assisted development.

Italian (Native) • **English** Professional working proficiency (Cambridge First Certificate)